

DTSC 5840

# Independent Study

|                                  |                                |               |
|----------------------------------|--------------------------------|---------------|
| FORMAT                           | CATEGORY                       | CREDITS       |
| On-Campus                        | Elective                       | None          |
| SECTION                          | BEST ROLE MATCH                | AVG ALIGNMENT |
| Section 4 – Elective (On-Campus) | Applied Scientist (12.5 / 100) | 5.1 / 100     |

5

CURRENT AVG SCORE  
across 6 job roles

71

POTENTIAL SCORE  
with recommended additions

+82

MAX SCORE GAIN  
best single-role improvement

## CURRENT COURSE DESCRIPTION

Subjects arranged individually through graduate advisor and faculty approval.

## INDUSTRY ALIGNMENT BY JOB ROLE

| JOB ROLE                  | CURRENT SCORE               | PROJECTED SCORE             | POINT GAIN | PRIORITY |
|---------------------------|-----------------------------|-----------------------------|------------|----------|
| Applied Scientist         | <div><div></div></div> 12.5 | <div><div></div></div> 94.3 | +81.8      | HIGH     |
| ML Engineer               | <div><div></div></div> 9.4  | <div><div></div></div> 52.1 | +42.6      | HIGH     |
| MLOps Engineer            | <div><div></div></div> 5.5  | <div><div></div></div> 62.6 | +57.1      | HIGH     |
| Machine Learning Engineer | <div><div></div></div> 3.4  | <div><div></div></div> 69.6 | +66.2      | HIGH     |
| AI Engineer               | <div><div></div></div> 0.0  | <div><div></div></div> 67.4 | +67.4      | HIGH     |
| Data Scientist            | <div><div></div></div> 0.0  | <div><div></div></div> 79.5 | +79.5      | HIGH     |

## RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold ( **example** ) appear as gaps across 3 or more job roles.

System Design

2 of 6 roles

systems

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

- 1

MEDIUM

Consider System Design content — relevant to 2 industry roles. Could be introduced via case studies or optional readings. Key terms: systems.

GAP KEYWORDS BY ROLE

**Applied Scientist** 12.5 → 94.3 (+81.8 pts)

lab computational research problems models technical data learning

**ML Engineer** 9.4 → 52.1 (+42.6 pts)

media data learning machine systems models model engineering

**MLOps Engineer** 5.5 → 62.6 (+57.1 pts)

detection data model systems learning machine media image

**Machine Learning Engineer** 3.4 → 69.6 (+66.2 pts)

data predictive engineering learning tools protection design development

**AI Engineer** 0.0 → 67.4 (+67.4 pts)

data engineering models learning build software technical code

**Data Scientist** 0.0 → 79.5 (+79.5 pts)

data different analytical social build engineering people problems

**Methodology:** Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum–Industry Alignment project team.